

ORIGINAL ARTICLE

Crop Management

A historical assessment of the number of days required to plant the Arkansas soybean crop

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Abstract

The primary objective of this paper is to quantitatively analyze the number of acres planted per suitable fieldwork day and the total number of fieldwork days available for the complete planting of soybeans (*Glycine max* L.) in Arkansas. Total days to plant soybeans in each spring from 1980 to 2023 averaged about 33 days, indicating that roughly 33 days are expected, on average, from the month of April to May based on USDA-NASS weekly crop progress and condition data. Furthermore, the average soybean planted acres is estimated at 106,736 acres per day within the study period. Recently, variations in this range appear more pronounced, likely due to the significant impact of extreme spring weather conditions. However, technological advancements have enabled soybean producers to plant crops as early as the first half of March or as late as June, contributing to the increased variability observed from year to year in recent times. Furthermore, Arkansas's optimal soybean planting window typically provides an average of 4.7 fieldwork days per week. These quantified estimates offer historical data on Arkansas's soybean planting performance and will prove highly valuable for informing future soybean planting strategies.

Plain Language Summary

Extreme weather events, like excessive rainfall and cooler-than-average temperatures, can reduce the number of days available to plant soybeans and delay soybean planting, resulting in lower-than-optimal yields and reduced returns. This study calculated the historical number of acres planted per day and the historical number of days available for planting the soybean crop in Arkansas using Arkansas's Crop Progress and Condition Report data from 1980 to 2023. The study found the average maximum number of acres planted per day is 106,736 acres and the average minimum number of days required to plant the Arkansas soybean crop is 33 days. The average historical number of suitable fieldwork days available per week to plant the

Abbreviations: ESPS, early soybean production system; MG, maturity group; PD, planting date.

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soybean crop within the optimal soybean planting window for Arkansas (roughly the last week of March through the first week of June) is 4.7 days. A soybean producer can use this information to estimate the number of days available to complete a soybean planting in years when planting has been delayed. Also, the methods used in this study may be applied to evaluate planting performance of other crops grown in other states or regions of the United States.

1 | INTRODUCTION

In 2023, Arkansas's soybean farmers produced 159.3 million bushels of soybean (*Glycine max* L.) crop, with an estimated production value is \$2.08 billion, and the state is ranked 10th in soybean production (USDA-NASS, 2024). The state is also ranked 10th in soybean export, valued around \$1.27 billion in 2022 (USDA-ERS, 2023). Thus, soybeans production, especially spring planting performance, plays an important role in supplying not only the state's soybean demand but also the demand outside of state, which all greatly contributes to the state's economy.

Despite strong soybean production in Arkansas, the spring planting season does not always progress as planned due to extreme weather (excessive rainfall, cooler than average temperatures), and the frequency of extreme weather events will most likely increase in the future (Rosenzweig et al., 2002). Consequently, these weather-related events led to high possibilities of poor crop yield potential and quality (Urban et al., 2015). A recent example occurred in 2019, when only 21% of Arkansas's total soybean acreage was planted by the second week of May 2019 due to extremely wet conditions with record-breaking rainfall of 15.8 inches above the historical average (NOAA, National Weather Service, 2019). This extremely wet year hindered planting progress, delaying not only soybean planting but also the planting of all other major summer crops grown in the state.

The soybean planted area in Arkansas has undergone significant transformations throughout the years. The highest soybean planted area occurred in 1979, reaching 5.2 million acres (Figure 1). A sharp decline from 1980 until 1990 occurred, then remained steady at around 3.2 million acres, with a standard deviation of 0.2 million acres from 1990 to 2023. Trend estimations for both irrigated and non-irrigated soybean acres show irrigated area steadily expanded by about 59,522.9 acres each year from 1972 to 2018, while non-irrigated acres declined roughly by 91,808.0 acres each year during the same period.

A major factor for the upward trend in irrigated soybean acreage in Arkansas is increased world demand for soybeans, particularly in China. Increased world demand for soybeans during this period increased the value of soybeans relative to other crops grown in the state, such as rice (*Oryza sativa*

L.) and cotton (*Gossypium hirsutum* L.) (Gautam & Watkins, 2021). More importantly, Arkansas's producers grow soybeans under irrigation to gain consistent and improved yields. The average soybean yield is 12.6 bu acre⁻¹ greater under irrigation relative to non-irrigated production. Today, roughly 83% of Arkansas's soybean planted area is irrigated (Watkins, 2023).

In the early 2000s, about 25% of total soybean planted area in Arkansas was planted using the Early Soybean Production System (ESPS) (Heatherly, 2005). Results from an earlier study suggested that southern irrigated farmland with ESPS can be planted as early as the first week of April, and northern region with irrigation can be planted as early as the second week of April under Maturity Group (MG) IV (Ashlock et al., 1998). The ESPS (as opposed to the full season production system where MG V-VII varieties are planted between May and June) improves yield and stability, mainly by avoiding heat and drought stress during seed fill and late season disease and insects' pressure (Moseley et al., 2022). Salmerón et al. (2016) studied the influence of planting dates (PD), latitude, and environmental factors on MG cultivar selection in soybean production under full irrigation from 2012 to 2014. Among MGs III, IV, V, and VI, cultivars consistently had the largest yield response as measured by percent relative yield across various planting dates from April 1 to June 15. Based on the soybean yield analysis across different MG, the planting date between April 1 and June 15 is a reasonable planting window that they used in their analysis.

Arkansas's agronomists and Extension educators recommend early planting as one pest management strategy to protect plant health and growth, particularly from fall armyworm (*Spodoptera frugiperda*) and Stink bugs (*Halyomorpha halys*) late in the season (Smith et al., 2009). Over 70% of insecticide applications are for fields in planting dates between June 1 and July 10 (Lorenz, 2017). A recent regional soybean study by Bateman et al. (2020) confirmed that planting before mid-May resulted in higher yields than planting after mid-May.

On the other hand, seed treatments such as inoculants increase soybean yields, especially in later planted fields. According to Roberts et al. (2016), inoculant seed treatments need to be applied to soybean fields planted after May 15. A wide range of options that are available due to technologies

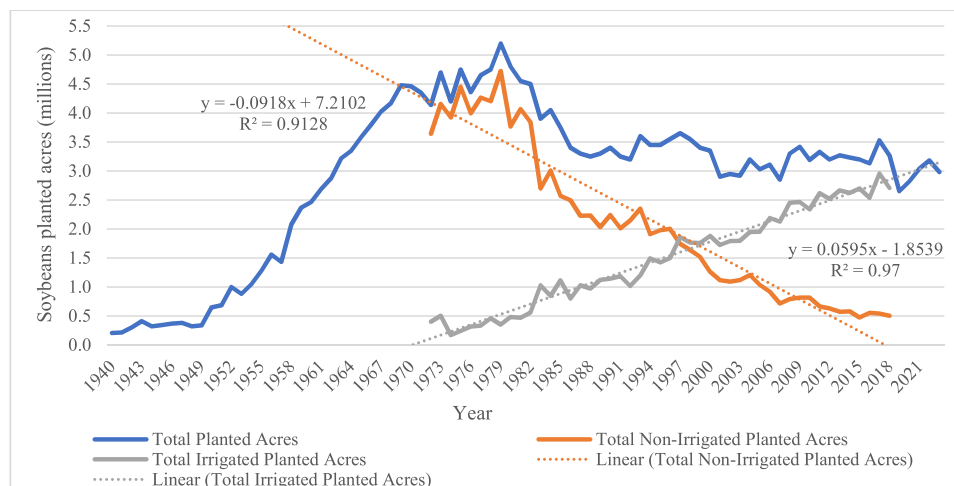


FIGURE 1 The historical total soybean planted acres from 1940 to 2023, total irrigated soybean planted acres, and total non-irrigated soybean planted acres between 1972 and 2018 in Arkansas. Planted acres data were sourced from USDA-NASS (2023a). USDA-NASS did not split soybean acres between irrigated and non-irrigated prior to 1972 and discontinued this split starting in 2019 due to lower funding availability (USDA-NASS, 2020).

(e.g., seed treatment, varieties, and irrigation) and different MGs make the soybean planting window a bit wider and more flexible relative to other crops such as rice and cotton in Arkansas. Despite the variety of technologies offered for planting soybeans, additional information is needed on the historical planting progress for the soybean crop in Arkansas.

Suitable fieldwork days for soybeans have been evaluated in the past for specific regions and time periods. Griffin and Kelley (2011) analyzed Arkansas's suitable fieldwork days from 1975 to 2009. On average, they found approximately 28.3 suitable fieldwork days for soybean planting in Arkansas between May 1 and June 12 during this period. Moreover, the authors found that soybean producers had about a 35% chance of having between 26 to 29 days available within these calendar dates. However, producers certainly need more than 28.3 days to plant soybeans in Arkansas due to (1) the massive expansion of soybean production in response to soybean demand, (2) constraints by planting equipment and labor capacity in the small planting window, and (3) recent highly variability weather variables during the spring planting season and extreme events in recent years. Since the 2011 study, spring weather has become more volatile, and soybean fieldwork planting day probabilities need to be updated to provide more accurate information for planting the Arkansas's soybean crop.

Massey et al. (2022) conducted a study in Missouri spanning from 1977 to 2017, focusing on suitable fieldwork days for planting and harvesting. This research quantified the probabilities of having specific numbers of fieldwork days available from April to May and September to October. Based on their state averages, about 25 suitable fieldwork days are available to plant soybeans in Missouri between May 1 and

June 12. Similarly, Shockley and Mark (2017) analyzed Kentucky's suitable fieldwork days from 1996 to 2016. Their study estimated percentiles for different criteria (15% for a bad year, 50% for a median year, and 85% for a good year) regarding suitable fieldwork days. They also provided graphical comparisons of the estimated fieldwork days suitable for corn (*Zea mays* L.) and soybean crops in Kentucky for these different scenarios. The goal was to assist Kentucky crop producers in evaluating their machinery capacities based on varying conditions. Using the 50th percentile probability of days suitable calculation, almost 29.5 suitable fieldwork days are available to plant soybeans in Kentucky from May 1 to June 12.

A recent study by Ward et al. (2020) found that Arkansas's peanut (*Arachis hypogaea*) producers will have about 29 suitable fieldwork days at the 50th percentile on average based on planting in the spring during the years of 1995 through 2018. In addition, Arkansas's cotton planting progress was estimated in 2017 by Griffin and Barnes (2017) and they figured about 19 suitable fieldwork days would be the minimum required days out of one planting window period (total 28 days in one cotton planting window in Arkansas). Until this day, there has been no study evaluating Arkansas's soybean planting progress using suitable fieldwork days.

Nielsen (2022) evaluated the historical progress of Indiana state corn and soybean crops using crop and condition report data from USDA-NASS for the period 1997 to 2021. His conclusion was that Indiana's corn and soybean planting progress in terms of total acres per week has not changed much over the study period, even though fewer farmers with planters are much more efficient and larger than 25 years ago. Although not specifically stated in Neilson (2022), roughly 20 suitable fieldwork days would be required to plant

Indiana's soybean crop based on rough computations of Figure 2 in Nielsen (2022).

The objective of this study is to assess Arkansas's historical soybean planting performance. Specifically, we wish to investigate the speed of soybean planting by evaluating the number of suitable fieldwork days producers needed to plant the entire state's soybean crop, as well as assess the suitable fieldwork days available and the annual variability associated with planting soybeans.

2 | MATERIALS AND METHODS

The methods and procedures used in this study follow those used for the Arkansas rice crop by Badarch and Watkins (2023) and were initially developed by Irwin (2022) and Irwin and Hubbs (2018) to evaluate corn planting performance in three major Corn Belt states (Illinois, Indiana, and Iowa) as well as corn planting progress for the entire United States as represented by results from 17 states (Irwin & Hubbs, 2020). These methods were also used by Nielsen (2022) to evaluate the planting progress of both corn and soybean acres in Indiana.

2.1 | Data sources

For this study, three sets of data were collected from the period 1980 to 2023, focusing only on the soybean crop in Arkansas: (1) soybean planted acres data (USDA-NASS, 2023a), (2) USDA-NASS weekly crop progress and condition report data (USDA-NASS, 2023b), and (3) suitable fieldwork days, also referred to as "days suitable for fieldwork" (USDA, 2023).

2.2 | Calculation method

The following three estimations were used for assessing historical Arkansas's soybean planting progress and performance:

1. The maximum Arkansas's soybean acres planted per suitable fieldwork day for each year was estimated by (i) multiplying each week's soybean planting progress percentage for a given year by the total soybean acres planted each year and (ii) summing the two peak weekly acres and dividing them by their respective sum of suitable fieldwork days.
2. The minimum number of suitable fieldwork days required to plant the Arkansas's soybean crop each year were estimated by dividing total planted soybean acres by the estimated maximum soybean acres planted per suitable fieldwork day.
3. The frequency distribution of historical probabilities for the number of available suitable fieldwork days per week during week 13 (the last week of March) through week 22 (the first week of June) planting window in Arkansas was estimated. The planting window was based on the upper and lower range of optimal soybean planting dates reported for eastern Arkansas in Salmerón et al. (2016).

3 | RESULTS AND DISCUSSION

Descriptive statistics of Arkansas's soybean planting progress are presented by period from 1980 to 2023 (Table 1). The number of suitable fieldwork days per week for planting soybeans on average for the period 2014–2023 were slightly smaller than in previous periods (first column in Table 1). The slightly fewer number of days is likely attributed to recent spring weather events like excessive rainfall in some of the years, for instance, 2023, 2020, 2019, 2017, 2016, and 2015, in which all had 20 inches or more of precipitation between March and June, leading to extremely wet spring conditions based on the National Weather Service and National Oceanic and Atmospheric Administration (NOAA) online weather database (NOAA, National Weather Service, 2023).

The mean percentage of soybeans planted by the second week of May exhibits a clear and consistent upward trend over time, particularly in recent years. Before the early 1990s, the planted area of the soybean crop in Arkansas before the second week of May was minimal, accounting for less than 10% of the state's total soybean planted area. However, after that time, there has been a remarkable progression in early-season soybean planting, with an approximate annual increase of 10 to 15% in each 10-year increment. Furthermore, this progression indicates increased planting efficiency over time. As mentioned earlier, the introduction of the ESPS in the mid-1990s in the southern United States played a significant role in this development (Moseley et al., 2022), aligning perfectly with the findings of this study.

Interestingly, despite the notable increase in early planting, the total soybean acreage in Arkansas has steadily declined, decreasing from slightly over 4.4 million acres in the 1980s to 3.1 million acres in recent years. Much of this reduction can be attributed to removal of marginal soybean area and the conversion of non-irrigated soybean area to irrigated soybean area. Irrigation allows farmers to produce more soybeans on fewer acres due to higher and more consistent yields under irrigation relative to non-irrigation (Watkins, 2023). According to the Farm Legislation report by Crowder and Davison (1990), Arkansas was one of the three leading states in double-cropped soybean acres planting during the 1970s and early 1980s; however, Arkansas's double-cropped soybean acres declined sharply due to lower soybean market price and other crops that supported by federal programs and their incentives.

TABLE 1 Descriptive statistics for Arkansas's soybean planting progress, 1980–2023.

Period	Mean			Standard Deviation			Minimum			Maximum		
	Planting days/week	% area planted by second week of May ^a	Annual area planted Acre (million)	Planting days/week	% area planted by second week of May ^a	Annual area planted Acre (million)	Planting days/week	% area planted by second week of May ^a	Annual area planted Acre (million)	Planting days/week	% area planted by second week of May ^a	Annual area planted Acre (million)
1980–1983	4.29	9.25	4.44	1.12	3.34	0.33	2.00	4.00	3.90	6.00	13.00	4.80
1984–1993	4.82	7.20	3.45	1.35	3.54	0.26	2.00	1.00	3.20	7.00	13.00	4.05
1994–2003	4.86	22.30	3.32	1.14	9.60	0.27	2.00	7.00	2.90	7.00	44.00	3.65
2004–2013	4.85	40.10	3.19	1.44	15.73	0.15	1.00	19.00	2.85	7.00	71.00	3.42
2014–2023	3.91	51.70	3.10	1.30	14.73	0.24	1.00	21.00	2.65	7.00	71.00	3.53
Overall	4.58	28.41	3.37	1.36	20.71	0.44	1.00	1.00	2.65	7.00	71.00	4.80

^aRepresents the week when 50% of soybean planted area was achieved on average during the decade 2014–2023.

The maximum soybean acres planted per day in Arkansas spanning from 1980 to 2023 are displayed in Figure 2. There is noticeable year-to-year volatility in soybean planted acres in the past 44 years. The average range roughly falls between 80,000 and 120,000 acres per day, except for a few years when significantly higher numbers of acres per suitable field-work day were planted. Notably, the early 1980s and 2005, 2006, 2007, and 2010 witnessed maximum average soybean acres planted per day, with values exceeding 120,000 acres. This was likely attributed to more suitable days for planting the crop in the timely fashion based on favorable weather. The year 2019 was recorded as the lowest maximum soybean planted acres per day, with roughly 60,000 acres statewide. This was primarily due to frequent rain events during the 2019 growing season (NOAA, National Weather Service, 2019).

Examining the graph, the trend for maximum planted soybean acres in Arkansas appears to show a slight overall decline. This decline is likely due to removal of marginal soybean areas, particularly from double cropped soybean acres declines starting from the late 1980s.

For the purpose of investigating the structural changes in the historical maximum soybean planting acres in Arkansas, the structural break Chow test (with 99% confidence bands around the null) was used in the study as displayed in Figure 3 (Fretheim, 2018). The test results indicated the cumulative sum plots drifted outside the confidence band between 1980s and the late 2000s, suggesting there is a structural break in this time series. The test value of 1.49 is greater than 1% critical value of 1.14, rejecting the null and indicates that the maximum planted soybean acres were not constant with mean acres in the study period. When excluding the years before the 1990s, the trend does not exhibit a pronounced downward trajectory, but the variability range does show significantly. While maximum values vary annually, the average figure remains at 106,736 acres per day from Week 13 through Week 22.

The minimum number of days required to complete the planting of Arkansas's soybean crop each year from 1980 to 2023 are displayed in Figure 4. In essence, the information in Figure 4 provides a response to our inquiry regarding the typical timeframe soybean producers take to plant the crop in Arkansas. With this evaluation, about 33 days are needed on average to complete the entire planting process; however, this historical graph does show significant year-to-year variation. Days above 40 signify a longer planting period attributable to uncontrollable factors like cold spring temperature or wet spring conditions. The higher number of days are associated with lower area planted, and is consistent throughout the years 2000, 2002, 2008, 2011, 2012, and 2019.

Our figure of 33 days on average to complete soybean planting in Arkansas is larger than the average found by Badarch and Watkins (2023) to complete rice planting in Arkansas (23 days), larger than the average number of days reported

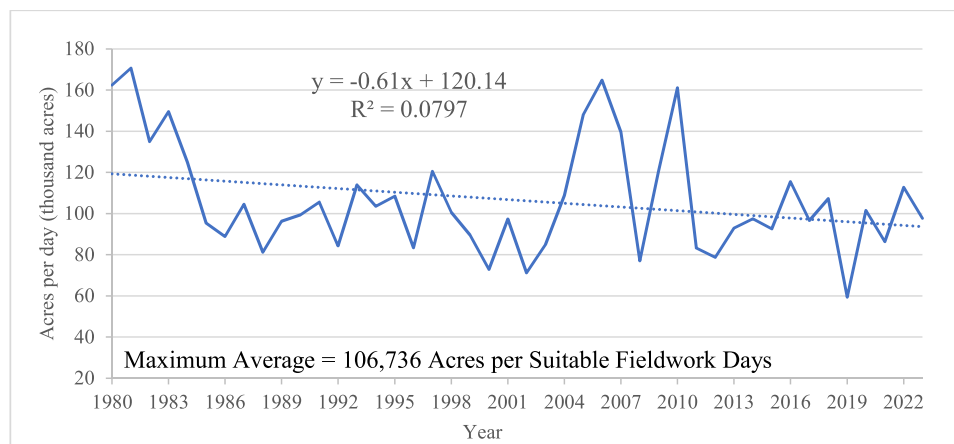


FIGURE 2 Maximum acres of soybean planted per day in Arkansas, an average of two peak weeks, 1980–2023.

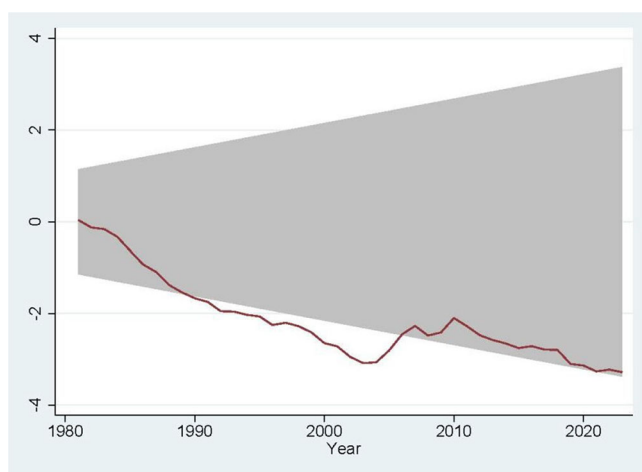


FIGURE 3 Structural break Chow test (with 99% confidence bands around the null) in maximum acres of soybean planted per day in Arkansas, an average of two peak weeks, 1980–2023.

by Irwin and Hubbs (2020) to complete planting of the Corn Belt (14 days), and larger than the rough computation of 20 days to plant soybean crops in Indiana based on Figure 2 of the Nielsen study (Nielsen, 2022). Griffin and Kelley (2011) reported 28.3 suitable fieldwork days available for planting soybeans, which is closer to our number, but their estimation procedure was a bit different, using a set planting window (May 1–June 12) rather than the methods employed in this study.

Primarily, our average number of 33 days is larger because of the diversity of crops grown in Arkansas. Furthermore, soybeans compete for timing of planting with rice, corn, and cotton in the state, and the soybean planting window is stretched out due to other crops being planted first. This is made clearly by Figure 5, which presents the 10-year average planting progress for the four major crops grown in Arkansas for the period 2013–2022. The curve for percent soybean planted area by week is much wider than curves for the other

three crops. Most rice and corn acres are planted before soybeans, whereas a significant amount of soybean acres are planted after cotton. Furthermore, roughly 20% of soybean acres in the state remain to be planted after planting concludes for the other three crops. These acres are planted during Weeks 23–28 (the second week of June through mid-July). Therefore, these acres are subject to lower than optimal yields and lower gross returns due to late planting.

The historical distribution of suitable fieldwork days per week for the planting window between the first week of March through the first week of June is presented for the period 1980–2023 in Figure 6. A broad spectrum in the number of suitable fieldwork days available per week mirrors the substantial variability in weather conditions within this specific time frame. Furthermore, soybean producers have a 21% chance (approximately 1 year out of 5) of experiencing only one, two, or three suitable fieldwork days per week. The most likely number of suitable fieldwork days per week is 4 (27% probability or 1 in 4 years). The average number of suitable fieldwork days in Arkansas between the last week of March and the first week of June is 4.7 days.

4 | SUMMARY AND DISCUSSIONS

This study conducted an in-depth analysis of Arkansas's historical soybean planting data spanning from 1980 to 2023 using publicly available crop progress and condition reports and annual planted soybean acres data supplied by USDA-NASS. The findings offer valuable and informative insights for Arkansas's soybean producers while also providing a historical context for the state's soybean planting performance. Specifically, the following key takeaways are listed: (1) the maximum planted soybean acres in 1 day during the planting season in Arkansas has remained relatively consistent over the past 44 years, with an average maximum of approximately 106,736 acres, although there is tremendous annual

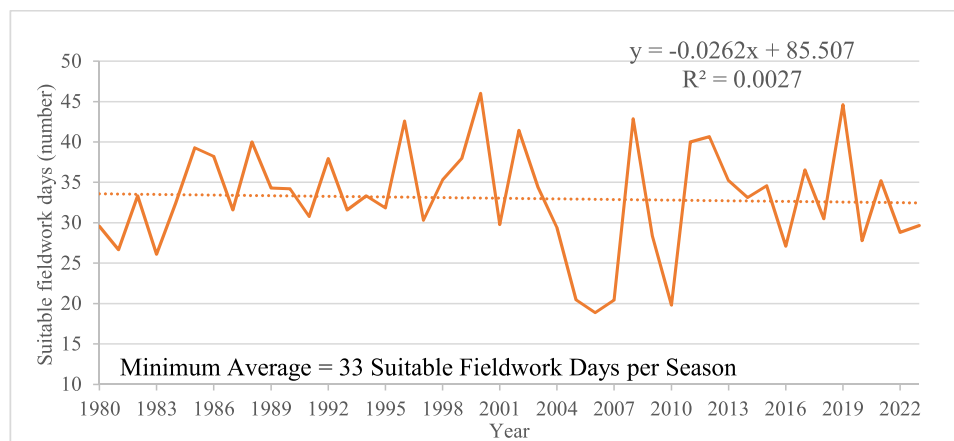


FIGURE 4 Minimum number of suitable fieldwork days for planting soybeans in Arkansas, 1980–2023.

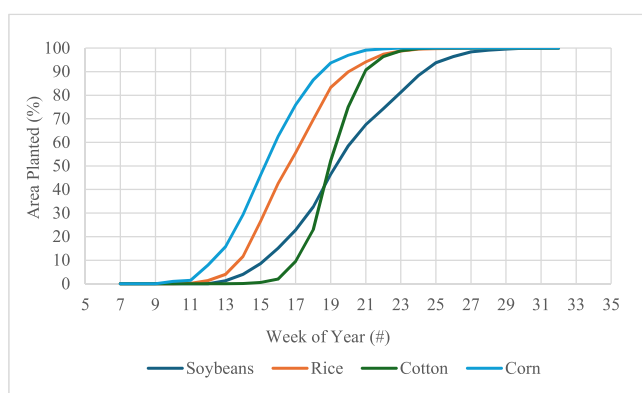


FIGURE 5 10-year average planting progress for soybeans, rice, cotton, and corn in Arkansas, 2013–2022 (USDA, 2023).

variability captured in the past, ranging from 80,000 to 120,000 acres. (2) the average minimum number of days required to complete soybean planting in Arkansas is 33 days, and ranges between 25 and 40 days. Variations in this range seem to be more exaggerated in recent years (2014–2023) due to extremely wet spring weather conditions. However, with more technology available for soybean planting, soybean producers were still able to plant half of their crop by the second week of May. Overall, no discernible linear trend is evident, indicating a stable requirement of around 33 days. (3) the weekly average number of suitable fieldwork days for the planting window between the last week of March and the first week of June is 4.7 days. The highest probabilities are associated with having four, five, or six suitable fieldwork days out of seven (72%, or approximately 3 out of 4 years), meaning that Arkansas's soybean producers can reasonably expect to plant soybeans in a timely manner in most years. The estimated planting time figures for this study thus provide valuable information for soybean producers in Arkansas.

4.1 | Farm management implication

We find these figures are useful for the completion of crop planting before important due dates. For instance, assume that a producer has not completed soybean planting by May 25 and still has significant area left to plant: the producer would have 2 weeks to complete planting within the optimal planting window (between April 1 and June 15) as used in the recent soybean study by Salmerón et al. (2016) in Arkansas. Based on the historical planting progress data, the expected suitable fieldwork days available per week is 4.7 on average, illustrated in Figure 5. Thus, the producer may expect to have on average 9.4 more days to finish planting soybeans within the optimal planting window; however, under a worst-case scenario, the farmer may have 6 or fewer days available to complete planting within the optimal planting window. If the producer needs more time than this to complete planting, the producer must prioritize which acres need to be planted first and which may be planted after the optimal planting date. In addition, this has implications on crop insurance coverage. In Arkansas, the producers must plant their soybean crops by June 15 as the final date to receive 100% of revenue guarantee, and by July 10 (the end date of the late planting period) to receive 75% of the revenue guarantee. Any area planted after July 10 is ineligible for the crop insurance coverage, but if the producers unable to plant soybeans during the late planting period due to excessive rainfall or flooding, the producer may receive prevented planting coverage, which covers about 60% of original coverage.

This study also provides evidence that annual soybean revenues in Arkansas could potentially be greater (and more producers could potentially receive greater profitability) if more soybean acres were planted in a timelier manner. Soybeans compete for planting time with other major crops grown in the state (rice, corn, and cotton) and tend to be the last crop planted. A significant portion of the states' soybean acres (roughly 20%) is planted after June 15 and is subject to

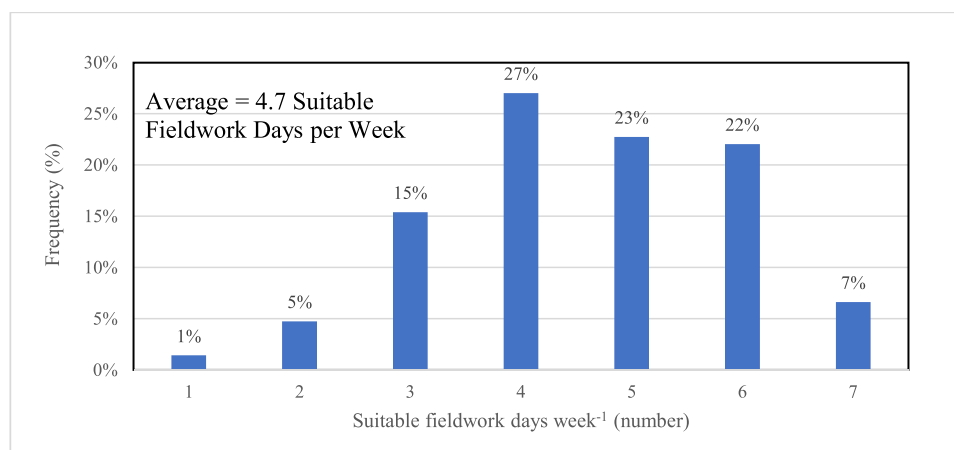


FIGURE 6 Total number of days available for planting soybeans in Arkansas during the spring planting window (last week of March through first week of June), 1980–2023.

lower-than-optimal yields based on soybean planting progress data for the last decade. Figure 2 of Salmerón et al. (2016) indicates that soybean yields for MG IVs decline significantly (relative yields below 80%) if planted after June 1 in Southeast Arkansas (the Rohwer location) and after June 15 in Northeast Arkansas (the Keiser location). In both locations, optimal MG IV yields are achieved for soybeans planted in late April to early May. Many soybean producers in Arkansas may be “leaving money on the table” by planting soybeans late.

While this study focused on Arkansas’s soybean planting performance, the state also cultivates other dominant field crops such as cotton, and corn. Future efforts to estimate suitable fieldwork days for these crops could prove worthwhile. Similar studies have been conducted in other states with their respective prominent crops. Hence, other states can apply similar estimation methods to gain insights into their crop planting historical perspectives. Lastly, this research addresses an information gap by providing quantified information sourced from USDA crop progress and condition reports on Arkansas’s soybean crop, offering valuable insights into the historical agricultural progress and performance of soybean planting in the state. Without generous support and tremendous data collection efforts by USDA-NASS, none of this research would be reality; thus, the study is dedicated to all the people working for USDA-NASS.

AUTHOR CONTRIBUTIONS

Bayarbat Badarch: Conceptualization; project administration; writing—original draft; writing—review and editing. **Bradley Watkins:** Funding acquisition; supervision; validation; visualization; writing—review and editing. **Terry Griffin:** Resources; supervision; visualization; writing—review and editing.

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CONFLICT OF INTEREST STATEMENT

The authors of this manuscript have no conflict interest to disclose.

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